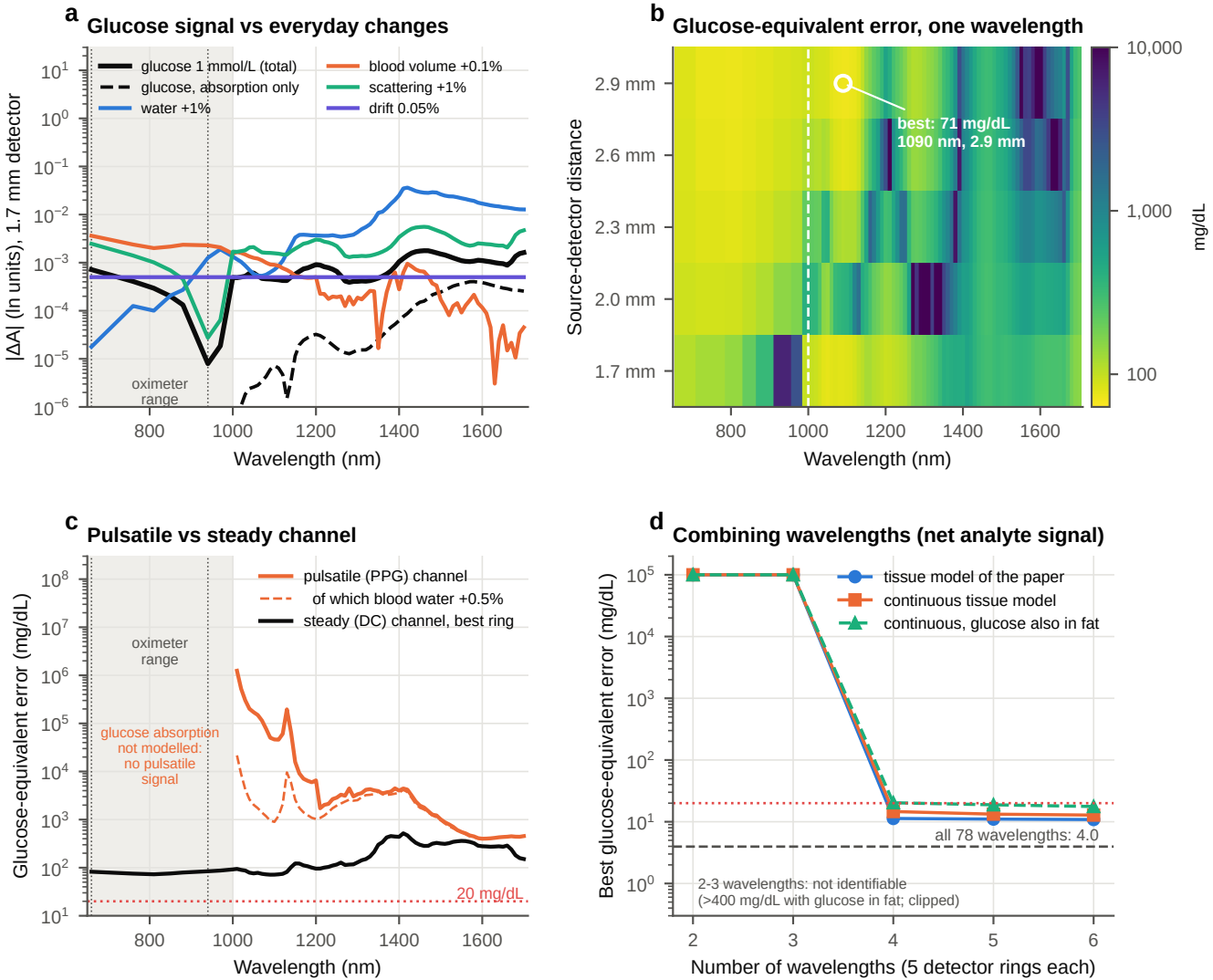
a
Glucose signal vs everyday changes
glucose 1 mmol/L (total)
glucose, absorption only
water +1%
blood volume +0.1%
scattering +1%
drift 0.05%
|ΔA| (ln units), 1.7 mm detector
oximeter range
Wavelength (nm)
b
Glucose-equivalent error, one wavelength
best: 71 mg/dL
1090 nm, 2.9 mm
Source-detector distance
2.9 mm
2.6 mm
2.3 mm
2.0 mm
1.7 mm
10,000
1,000
100
mg/dL
Wavelength (nm)
c
Pulsatile vs steady channel
pulsatile (PPG) channel
of which blood water +0.5%
steady (DC) channel, best ring
oximeter range
glucose absorption not modelled: no pulsatile signal
20 mg/dL
Glucose-equivalent error (mg/dL)
Wavelength (nm)
d
Combining wavelengths (net analyte signal)
tissue model of the paper
continuous tissue model
continuous, glucose also in fat
all 78 wavelengths: 4.0
2-3 wavelengths: not identifiable
(>400 mg/dL with glucose in fat; clipped)
Best glucose-equivalent error (mg/dL)
Number of wavelengths (5 detector rings each)